AI CAREER & COURSE RECOMMENDATION SYSTEM

Project Report

PRAJJWAL PANDEY

VIT BHOPAL UNIVERSITY
FUNDAMENTALS IN AI & ML

◈ 1. ABSTRACT

This project presents an **AI-based Career and Course Recommendation System** developed using a rule-based approach. The system helps users select suitable career paths based on their interests and skills such as mathematical ability, communication, and creativity.

The system simulates an **expert decision-making process** using predefined rules and knowledge representation. It provides not only career suggestions, but also relevant courses required to achieve those careers.

The application is built using **Python and Tkinter**, offering a user-friendly graphical interface. The system covers multiple domains including technology, business, healthcare, agriculture, and more, making it practical and scalable.

◈ 2. INTRODUCTION

Artificial Intelligence (AI) is the simulation of human intelligence in machines. One important branch of AI is **Expert Systems**, which use knowledge and rules to make decisions similar to human experts.

Choosing a career is one of the most important decisions in a student's life. Many students face confusion due to lack of proper guidance. This project aims to solve this problem by developing a **rule-based recommendation system**.

The system analyzes user inputs and suggests suitable careers along with courses, making the decision process easier and more structured.

◈ 3. PROBLEM STATEMENT

Students often face difficulty in choosing the right career path due to:

- Lack of proper guidance
- Confusion about interests and skills
- Limited access to career counseling

This project aims to develop a system that:

- Provides career recommendations
- Suggests relevant courses
- Uses logical decision-making

◈ 4. OBJECTIVES

The main objectives of this project are:

- To design a rule-based AI system
- To recommend careers based on user inputs
- To suggest courses required for selected careers
- To create a user-friendly graphical interface
- To apply AI concepts like knowledge representation

◈ 5. LITERATURE REVIEW

Existing systems for career guidance include:

- Online career counseling platforms
- Human career advisors
- Basic recommendation tools

Limitations:

- Expensive or not easily accessible
- Not personalized
- Lack of transparency in decision-making

This project overcomes these limitations by providing:

- Simple rule-based logic
- Transparent recommendations
- Easy-to-use interface

◈ 6. METHODOLOGY

The system follows a **rule-based approach**:

Step 1: Input

User provides:

- Interest
- Math skill
- Communication skill
- Creativity

Step 2: Processing

- Rules are applied using IF-ELSE conditions
- System matches input with predefined knowledge

Step 3: Output

- Career recommendations
- Course suggestions

This approach is similar to an **AI expert system**, where knowledge is stored and used for inference.

◈ 7. SYSTEM DESIGN

◆ Components:

- Input Interface (Tkinter GUI)
- Rule Engine
- Knowledge Base
- Output Display

◆ Working:

1. User selects inputs
2. System processes rules
3. Career is recommended
4. Courses are displayed

◈ 8. IMPLEMENTATION

The system is implemented using:

◈ Python

- Core programming logic

- Rule-based conditions

◈ Tkinter

- GUI development
- Input fields and output display

◈ Logic Example:

```
if interest == "coding" and math == "high":
    career = "Software Engineer"
```

The system contains multiple rules covering different domains like:

- Technology
- Business
- Agriculture
- Healthcare

```python
import tkinter as tk
from tkinter import ttk

# ---------- LOGIC ----------

def recommend_career(interest, math, communication, creativity):
    results = []

    # TECH
    if interest == "Coding" and math == "High":
        results.append(("Software Engineer", ["Python", "DSA", "Algorithms"]))

    if interest == "Coding" and communication in ["Medium", "High"]:
        results.append(("Data Scientist", ["Python", "ML", "Statistics"]))

    if interest == "Gaming":
        results.append(("Game Developer", ["Unity", "C#", "Game Design"]))

    # DESIGN
    if interest == "Design" and creativity == "High":
        results.append(("UI/UX Designer", ["Figma", "UX Design"]))

    # BUSINESS
    if interest == "Management":
```

◈ 9. FEATURES

- Modern GUI using Tkinter
- Rule-based AI system
- Supports 15+ interests

- Includes agriculture domain
- Provides course recommendations
- Fast and user-friendly

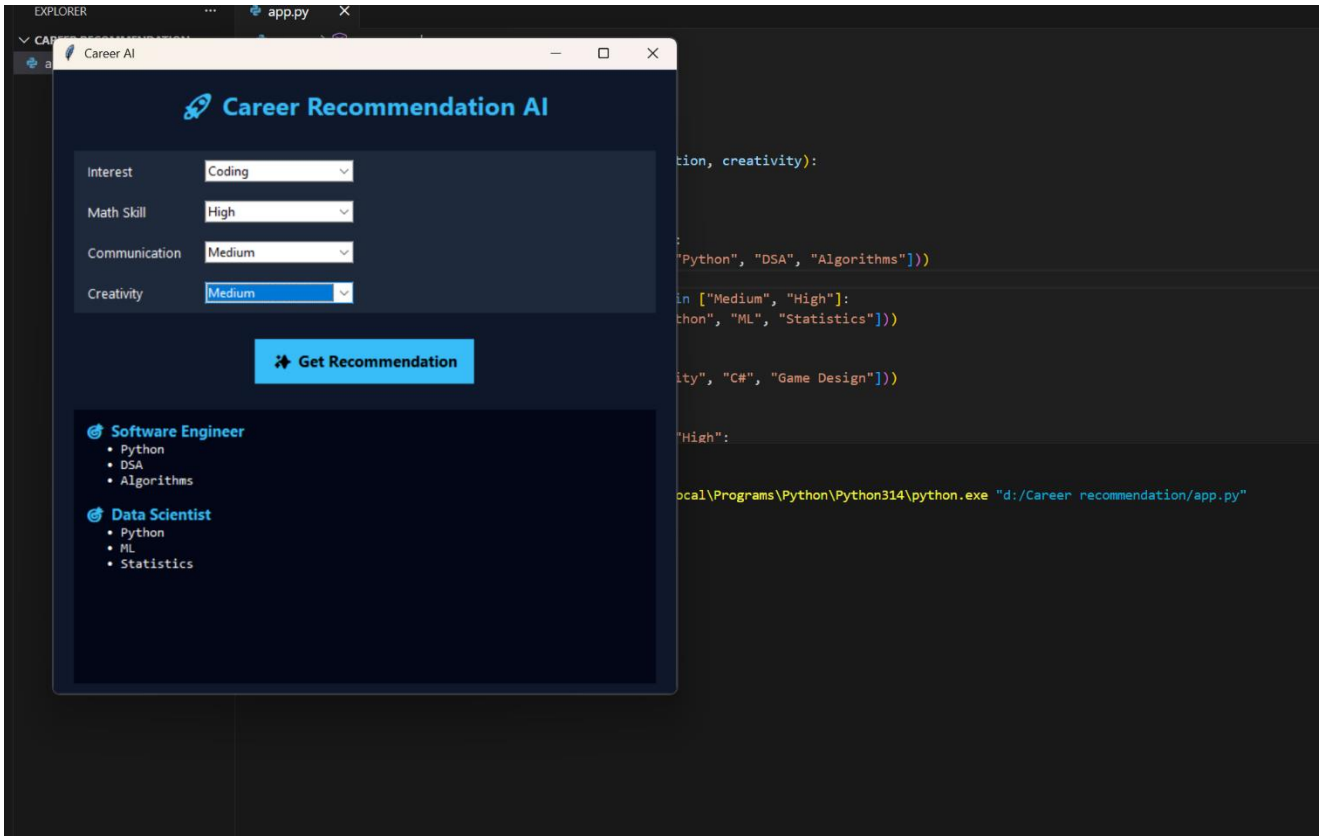
◈ 10. RESULTS / OUTPUT

The system successfully generates career recommendations.

Example Output:

Career: Software Engineer
Courses:
- Python
- Data Structures
- Algorithms

The GUI displays results clearly and is easy to use.



◈ 11. ADVANTAGES

- Easy to use
- Fast decision-making
- Covers multiple domains
- Low cost (no external tools required)
- Transparent logic

◈ 12. LIMITATIONS

- Rule-based (no learning capability)
- Limited to predefined rules
- Not personalized using real data

◈ 13. FUTURE SCOPE

The system can be improved by:

- Adding Machine Learning models
- Creating a web application
- Adding chatbot interface
- Using real-world datasets
- Providing more personalized recommendations

◈ 14. CONCLUSION

This project successfully demonstrates how **AI concepts like rule-based systems and knowledge representation** can be used to solve real-world problems such as career guidance.

The system provides accurate recommendations based on user inputs and offers a simple yet effective solution for students.

◈ 15. REFERENCES

- Python Official Documentation
- Tkinter Documentation
- AI & Expert System Tutorials
- Online educational resources